

PIOTTA, Switzerland

WMO: 067530

Lat: 46.5147N

Lon: 8.6881E

Elev: 1016

StdP: 89.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.6	-8.0	-19.2	0.8	-4.2	-16.1	1.0	-3.6	9.5	1.4	8.6	1.9	1.8	280	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	27.3	17.3	25.7	16.8	24.2	16.2	18.8	24.7	17.9	23.3	17.1	22.2	4.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.5	21.3	15.9	12.8	20.2	15.1	12.2	19.5	57.8	24.9	54.7	23.3	52.0	22.3	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.0	6.3	-12.0	29.7	1.9	2.1	-13.3	31.2	-14.4	32.4	-15.5	33.6	-16.8	35.1		
(5)			-12.8	20.3	2.0	1.1	-14.3	21.1	-15.4	21.8	-16.6	22.4	-18.0	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-1.2	0.4	4.5	8.0	12.0	16.1	17.9	17.2	13.3	9.1	3.5	-0.4		
	DBStd	7.43	3.37	3.54	3.64	3.35	3.12	3.23	2.64	2.66	2.84	3.04	3.36	3.42		
	HDD10.0	1460	346	268	174	78	15	1	0	0	5	53	197	323		
	HDD18.3	3684	604	501	431	311	197	81	39	52	153	287	446	582		
	CDD10.0	879	0	0	2	17	77	184	246	224	103	24	1	0		
	CDD18.3	62	0	0	0	0	1	14	27	18	1	0	0	0		
	CDH23.3	514	0	0	0	0	13	144	208	138	11	0	0	0		
	CDH26.7	78	0	0	0	0	1	22	28	26	1	0	0	0		
(14) Wind	WSAvg	2.4	2.2	2.5	2.7	2.8	2.8	2.8	2.8	2.5	2.3	1.9	2.0	2.0		
(15) Precipitation	PrecAvg	1594	91	92	95	133	164	156	139	158	155	138	172	101		
	PrecMax	2075	213	236	199	290	436	295	259	239	427	432	579	215		
	PrecMin	1099	23	22	13	21	52	60	65	49	24	19	27	10		
	PrecStd	249	49	54	50	72	93	60	51	46	98	89	125	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	12.5	17.6	21.6	25.0	29.0	29.3	29.7	24.7	19.7	14.1	10.8		
		MCWB	4.4	5.1	7.5	11.1	15.0	17.7	17.0	17.1	15.6	11.7	8.2	4.0		
	2%	DB	7.0	9.4	15.0	18.8	22.5	26.8	27.3	26.6	22.5	17.1	11.3	7.8		
		MCWB	2.9	3.6	6.6	9.7	13.2	16.7	17.3	17.8	15.3	11.3	6.7	3.1		
	5%	DB	5.1	7.3	12.7	16.6	20.5	24.9	25.7	24.6	20.3	15.4	9.6	5.9		
		MCWB	1.4	2.5	5.6	8.4	12.1	16.2	16.9	17.0	14.0	10.7	6.0	2.5		
	10%	DB	3.4	5.4	10.5	14.5	18.4	23.0	24.1	23.0	18.6	13.9	8.1	4.3		
		MCWB	0.6	1.5	4.7	7.4	11.2	15.3	16.2	16.2	13.2	10.5	5.6	1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	5.7	8.6	11.7	15.9	19.5	20.3	19.9	17.2	13.8	9.7	5.9		
		MCDB	8.4	11.2	15.2	19.8	22.6	26.1	26.1	25.9	22.3	16.3	11.9	8.1		
	2%	WB	3.4	4.1	7.3	10.3	14.3	18.0	18.9	18.6	16.0	12.9	8.0	4.3		
		MCDB	6.1	8.3	12.9	17.2	20.3	24.3	24.7	24.4	20.8	15.1	9.9	6.3		
	5%	WB	2.1	3.0	6.2	9.2	13.1	16.9	17.9	17.8	15.0	11.9	7.1	3.1		
		MCDB	4.1	6.2	11.3	14.7	18.6	23.1	23.2	22.8	18.9	14.2	8.5	5.0		
	10%	WB	1.2	2.1	5.2	8.2	12.1	15.9	17.1	16.9	14.1	11.0	6.2	2.1		
		MCDB	2.7	4.9	9.7	13.1	16.7	21.5	22.1	21.4	17.4	13.2	7.7	3.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.2	8.6	8.9	9.1	10.1	10.2	9.6	8.6	7.9	5.8	5.3		
		MCDBR	8.3	10.0	12.4	13.3	13.2	13.5	12.9	12.6	11.5	10.5	8.6	7.8		
	5% WB	MCWBR	5.3	5.9	6.3	6.3	6.4	6.3	5.9	6.0	6.0	5.9	5.3	4.9		
		MCWBR	6.7	8.8	10.7	11.3	11.1	12.0	11.3	11.0	9.9	8.3	6.7	5.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.270	0.301	0.354	0.411	0.431	0.442	0.436	0.424	0.393	0.359	0.307	0.267		
		taud	2.272	2.186	2.081	2.025	2.100	2.135	2.160	2.206	2.269	2.323	2.354	2.312		
	Ebn at Noon	Ebn at Noon	824	861	857	839	837	828	827	820	813	786	771	786		
		Edh at Noon	78	105	137	160	154	150	144	132	114	93	73	67		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.61	2.58	3.79	4.47	4.77	5.62	5.61	4.87	3.79	2.52	1.57	1.34		
	RadStd	RadStd	0.20	0.30	0.55	0.73	0.48	0.50	0.49	0.43	0.37	0.30	0.23	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	+0.83	+0.52	N/A	N/A	N/A	+58	N/A
(47)	Regional (120 neighbors)	N/A	N/A	N/A	+0.73	+0.42	+0.32	N/A	N/A	+57	+15

Nomenclature: See separate page